

Diego Pilares Gallego

Madrid, Spain

 dpilares01@gmail.com  linkedin.com/in/diegopilares/  dpilares01.github.io/  +34 689525579

ABOUT ME

MSc Motorsport Engineering student at Oxford Brookes with a background in automotive engineering, focused on vehicle dynamics, data analysis, and CAD. Driven by curiosity and a hands-on mindset, I enjoy solving engineering challenges and creating innovative, high-quality solutions. Passionate about motorsport, I am proactive, collaborative, and always seeking to deepen my understanding of performance and vehicle behaviour.

EDUCATION

MSc Motorsport Engineering

Sept. 2024 – Sept. 2025; Oxford, UK

Oxford Brookes University

- Expected mark: Distinction (pending dissertation)
- Key modules: Advanced Vehicle Dynamics - Advanced Vehicle Aerodynamics - Laptime Simulation and Race Engineering - Composite Design and Impact Modelling.

BEng Automotive Engineering

Sept. 2019 – Jan. 2025; Madrid, Spain

Universidad Francisco de Vitoria – Motor & Sport Institute

- Final Mark: 6.91/10 (2:1 UK equivalent)
- Key modules: Vehicle Dynamics - Design, Prototyping and Testing - Transmission and Gearbox - Suspension - Steering and Braking Systems – Powertrain – Aerodynamic - Fluid Dynamics – Thermodynamics - Machines and Mechanisms Theory - Advanced Materials.

PROFESSIONAL EXPERIENCE

Formula Student Team Member

Sept. 2024 – May 2025; Oxford, UK

Oxford Brookes Racing

- Collaborating with the Suspension & Structures departments to design and test new brake rotors and hub components.
- Using SolidWorks for CAD and HyperMesh for FEA, to enable structural and thermal analysis, as well as topology optimization.

Industrial Engineer Placement

Feb. 2024 – Sept. 2024; Madrid, Spain

NTT DATA

- Placement on Automotive department.
- Involved in the development of corrective measures and new functionalities.
- Produced technical documentation, training presentations, analysed and resolved issues for the portals of BMW Spain, Portugal, Italy, Mexico, Brazil and India. Created layouts for Hyundai Spain.

Formula Student Team Member

Sept. 2019 – July 2021; Madrid, Spain

UFV Racing

- Collaborated with the Electronics department, creating wiring diagrams and PCB layouts via LTSpice and EasyEDA, developed Aim PDM integration and engine mapping.
- Collaborated with the Systems and Testing department to install powertrain, steering, suspension and brake components onto the vehicles, CAD of components in SolidWorks, 3D printing of test components via Ultimaker Cura and TIG welding.

PROJECTS

Full project portfolio available at: dpilares01.github.io/

MSc Dissertation: Hydrogen-fuelled LTC/CAI-IC engine control strategy optimization

Oxford Brookes University

Jan. 2025 – Present

- Simulation of a 1D Low-Temperature Combustion engine using [GT-Suite](#).
- Control strategy development utilising DOE and optimisation using [GT-Suite](#)'s ML algorithms.
- Focused on hydrogen to maximise efficiency and reduce NOx emissions.

Next-gen F1 compliance and setup study

Apr. 2025 – May. 2025

Oxford Brookes University

- Modified a 2024-spec F1 vehicle geometry to meet 2026 FIA technical regulations using [AVL VSM](#).
- Simulated skid-pad and on-track performance and optimised suspension, tyres and brake balance.
- Used [AVL Drive Race](#) to analyse telemetry and validate setup improvements.

LMGT3 DrivAer front aero devices design and CFD validation

Oct. 2024 – Dec. 2024

Oxford Brookes University

- Designed front aero devices for a DrivAer model in [SolidWorks](#), following LMGT3 regulations.
- Performance CFD validation on [Star CCM+](#), assessing aerodynamic performance.
- Devised a wind tunnel experimental validation methodology.

Suspension Dynamics and Ride Optimisation

Oct. 2024 – Dec. 2024

Oxford Brookes University

- Bicycle model and Twin-Track model developed and analysed in [MSC Adams](#).
- Developed 1, 2 and 4 Degrees of Freedom suspension models, analysed and optimized across cornering and vertical load cases in [MSC Adams](#).

BEng Dissertation: Carbon fibre centre-lock rim for Formula Student

Aug. 2024 – Jan. 2025

Universidade Francisco de Vitoria

- Designed a lightweight carbon fibre rim with integrated centre-lock nut CAD model in [SolidWorks](#).
- Performed FEA structural analysis on [SolidWorks](#) and determined the materials using [Granta](#).
- Designed the rim and the centre-lock nut in compliance with ISO and Formula Student regulations.

Brake cooling duct design for race oriented GR86

Sep. 2023 – Jan. 2024

Universidade Francisco de Vitoria - Motor & Sport Institute

- Scanned a GR86 front geometry using [EXScan](#) for precise duct integration.
- Designed CAD model in [SolidWorks](#) and 3D printed multiple prototype designs with [Ultimaker Cura](#).
- Performed FEA for airflow study and cooling performance improvements in [SolidWorks](#).

SKILLS

- **Vehicle and systems simulation:** AVL VSM, GT-Suite, MSC Adams, MATLAB.
- **CAD, FEA and CFD:** SolidWorks, Star CCM+, XFLR5, Femap, Hypermesh.
- **Data acquisition and analysis:** AVL Drive Race, MoTeC i2.
- **Productivity:** Microsoft Office, Google Workspace.

LANGUAGES

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|---------------------------|--|
| • Spanish - Fluent | • English - Fluent |
| <i>Native</i> | <i>Pearson PTE Academic Overall Score 80 – 2024</i> |
| | <i>Cambridge English Certificate level B2 – 2019</i> |

INTERESTS

Motorsport Engineering: Deep interest in vehicle design, performance optimisation and race technology evolution.

Sim Racing & Go-Karting: Exploring setup theory, tyre behaviour and handling dynamics.

Retro Electronics Restoration: Repair and modification of video game consoles and other electronics through reverse engineering and creative problem solving.

Gym training: Committed to training routines to support focus, resilience and discipline.

REFERENCES AND EMPLOYMENT STATUS

References available upon request

Currently on a UK Student Visa until January 2026.